

5. MULTIPLE INTEGRALS

$$\int a dx = ax$$

$$\int x^n = \frac{x^{n+1}}{n+1}$$

$$\int e^{ax} = \frac{1}{a} e^{ax}$$

$$\int \sin x = -\cos x + C$$

$$\int \cos x = \sin x + C$$

$$\int \operatorname{cosec}^2 x dx = -\cot x + C$$

$$\int \sec^2 x dx = \tan x + C$$

$$\int \sec x \tan x dx = \sec x + C$$

$$\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{\sqrt{a^2 + x^2}} dx = \cos^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1}\left(\frac{x}{a}\right) + C$$

$$\int -\frac{1}{a^2 + x^2} dx = \frac{1}{a} \cot^{-1}\left(\frac{x}{a}\right) + C$$

$$\int \frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \operatorname{sech}^{-1}\left(\frac{x}{a}\right) + C$$

$$\int -\frac{1}{x\sqrt{x^2 - a^2}} dx = \frac{1}{a} \operatorname{cosech}^{-1}\left(\frac{x}{a}\right) + C$$

Q) Evaluate $\int_0^3 \int_1^2 xy(1+x+y) dy dx$

Sol) $\int_0^3 \int_1^2 xy + x^2y + xy^2 dy dx$

$$\int_0^3 \left[x \frac{y^2}{2} \Big|_1^2 + x^2 \frac{y^2}{2} \Big|_1^2 + x \frac{y^3}{3} \Big|_1^2 \right] dx$$

$$\int_0^3 \left[\frac{x}{2} (4-1) + \frac{x^2}{2} (4-1) + \frac{x}{3} (8-1) \right] dx$$

$$\int_0^3 \left[\frac{3}{2} x + \frac{3}{2} x^2 + \frac{7}{3} x \right] dx$$

$$\left[\frac{3}{2} \cdot \frac{x^2}{2} \Big|_0^3 + \frac{3}{2} \cdot \frac{x^3}{3} \Big|_0^3 + \frac{7}{3} \cdot \frac{x^2}{2} \Big|_0^3 \right]$$

$$\frac{3}{4} [9-0] + \frac{1}{2} [27-0] + \frac{7}{6} [9-0]$$

$$\frac{3}{4} \times 9 + \frac{1}{2} \times 27 + \frac{7}{6} \times 9$$

$$\frac{27}{4} + \frac{27}{2} + \frac{21}{2} = \frac{27+54+42}{4} = \frac{123}{4}$$

Q) Evaluate $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2+y^2) dx dy$

sol) $\int_0^a \int_0^{\sqrt{a^2-y^2}} (x^2+y^2) dx dy$

$$\int_0^a \left[\frac{x^3}{3} + y^2 x \right]_0^{\sqrt{a^2-y^2}} dy$$

$$\int_0^a \left[\frac{(a^2-y^2)^{3/2}}{3} + y^2 (a^2-y^2)^{1/2} \right] dy$$

$$I_2 = \int_{y=0}^a y^2 (a^2-y^2)^{1/2} dy$$

Put $y = a \sin \theta \Rightarrow dy = a \cos \theta d\theta$

limits : when $y=0 \Rightarrow \sin \theta = 0 \Rightarrow \theta = 0$

$y=a \Rightarrow a = a \sin \theta$

$\sin \theta = 1$

$\theta = \frac{\pi}{2}$

$$\int_0^{\pi/2} a^2 \sin^2 \theta (a^2 - a^2 \sin^2 \theta)^{1/2} a \cos \theta d\theta$$

$$\int_0^{\pi/2} a^4 \sin^2 \theta (1 - \sin^2 \theta)^{1/2} \cos \theta d\theta = \int_0^{\pi/2} a^4 \sin^2 \theta \cos^2 \theta d\theta$$

$$\int_0^{\pi/2} a^4 \frac{(\sin 2\theta)^2}{4} d\theta$$

$$I_2 = a^4 \cdot \frac{1}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = a^4 \frac{\pi}{16}$$

$$I = \int_0^a \frac{(a^2-y^2)^{3/2}}{3} \Rightarrow \frac{1}{3} \int_0^a (a^2-y^2)^{3/2} dy$$

$$= \frac{1}{3} \int_0^{\pi/2} (a^2 - a^2 \sin^2 \theta)^{3/2} a \cos \theta d\theta = \frac{1}{3} \int_0^{\pi/2} a^4 \cos^3 \theta \cos \theta d\theta$$

$$= \frac{a^4}{3} \int_0^{\pi/2} \cos^4 \theta d\theta = \frac{a^4}{3} \int_0^{\pi/2} \cos^2 \theta d\theta$$

$$\int_0^4 \int_{y^2/4}^y \frac{y}{x^2+y^2} dx dy$$

sol

$$\int_0^4 y \int_{y^2/4}^y \frac{1}{x^2+y^2} dy = \int_0^4 y \left[\frac{1}{y} \tan^{-1}\left(\frac{x}{y}\right) \right]_{y^2/4}^y dy$$

$$\int_0^4 \tan^{-1}\left(\frac{y}{y}\right) - \tan^{-1}\left(\frac{y^2/4}{y}\right) dy = \int_0^4 \tan^{-1}(1) - \tan^{-1}\left(\frac{y}{4}\right) dy$$

$$\int_0^4 \tan^{-1}(1) - \tan^{-1}\left(\frac{y}{4}\right) dy = \int_0^4 \frac{\pi}{4} - \tan^{-1}\left(\frac{y}{4}\right) dy$$

$$= \left[\frac{\pi}{4} \cdot y \right]_0^4 - \int_0^4 \tan^{-1}\left(\frac{y}{4}\right) \cdot 1 dy$$

$$= \frac{\pi}{4} \cdot y \Big|_0^4 - \int_0^4 \tan^{-1}\left(\frac{y}{4}\right) dy$$

$$= \pi - \left\{ \tan^{-1}\left(\frac{y}{4}\right) \int 1 dy - \int \left[\frac{1}{1 + \frac{y^2}{16}} \left(\frac{1}{4}\right) \cdot \int 1 dy \right] dy \right\}$$

$$= \pi - \left\{ \tan^{-1}\left(\frac{y}{4}\right) \cdot y - \int \frac{16}{16+y^2} \cdot \frac{1}{4} y dy \right\}$$

$$= \pi - \left\{ y \tan^{-1}\left(\frac{y}{4}\right) - 4 \int \frac{y}{16+y^2} dy \right\}$$

$$= \pi - \left[y \tan^{-1}\left(\frac{y}{4}\right) - 2 \int \frac{2y}{16+y^2} dy \right]$$

$$= \pi - \left[y \tan^{-1}\left(\frac{y}{4}\right) - 2 \log(16+y^2) \right]_0^4$$

$$= \pi - \left[4 \tan^{-1}\left(\frac{4}{4}\right) - 2 \log(16+4^2) \right] - [0 - 2 \log(16)]$$

$$= \pi - [4 \tan^{-1}(1) - 2 \log 32 + 2 \log 16]$$

$$= \pi - \frac{\pi}{4} + 2 \log 32 - 2 \log 16 = \pi - \pi + 2 \log \frac{32}{16} = 2 \log 2$$

Q) Evaluate $\iint_R y^2 dx dy$ where R is region bounded within the parabolas $y^2 = 4x$; $x^2 = 4y$

Sol) To find the point of intersection of two curves be simplified the curves as follows.

$$x^2 = 4y \rightarrow (1)$$

$$y^2 = 4x \rightarrow (2)$$

$$x = \frac{y^2}{4}$$

$$\left(\frac{y^2}{4}\right)^2 = 4y$$

$$\frac{y^4}{16} = 4y$$

$$y^4 = 64y$$

$$y^4 - 64y = 0$$

$$y[y^3 - 64] = 0$$

$$y = 0$$

$$x = 0$$

$$y = 4$$

$$x = \frac{4^2}{4}$$

$$\boxed{x = 4}$$

point of intersection $(0,0)$ $(4,4)$

limits: $x \rightarrow 0$ to 4

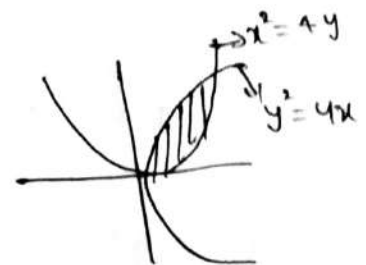
$$y^2 = 4x \Rightarrow y = 2\sqrt{x} \text{ (U.L.)}$$

$$4x^2 = 4y \Rightarrow y = \frac{x^2}{4} \text{ (L.L.)}$$

$$y \rightarrow \frac{x^2}{4} \text{ to } 2\sqrt{x}$$

$$\int_{x=0}^4 \int_{\frac{x^2}{4}}^{2\sqrt{x}} y^2 dy dx$$

$$\int_0^4 \left[\frac{y^3}{3} \right]_{\frac{x^2}{4}}^{2\sqrt{x}} dx = \int_0^4 \frac{1}{3} \left[(2\sqrt{x})^3 - \left(\frac{x^2}{4}\right)^3 \right] dx$$



$$\begin{aligned}
 &= \int_0^4 \frac{1}{3} \left\{ \left[8(x^{3/2}) - \frac{1}{64} x^6 \right] \right\} dx \\
 &= \frac{1}{3} \left\{ 8 \frac{x^{5/2}}{5/2} - \frac{1}{64} \frac{x^7}{7} \right\}_0^4 \\
 &= \frac{1}{3} \left\{ 8 \cdot \frac{2}{5} 4^{5/2} - \frac{1}{43} \frac{4^7}{7} \right\} = 0 \\
 &= \frac{1}{3} \left\{ \frac{16}{5} 2^5 - \frac{4^4}{7} \right\} \\
 &= \frac{16 \times 2^5}{15} - \frac{256}{21} \\
 &= 21.9
 \end{aligned}$$

Double Integral in polar coordinates

Q1 Evaluate $\int_0^\pi \int_0^{a \sin \theta} r dr d\theta$

sol

$$\int_{\theta=0}^\pi \int_{r=0}^{a \sin \theta} r dr d\theta = \int_0^\pi \left[\frac{r^2}{2} \right]_0^{a \sin \theta} d\theta = \frac{1}{2} \int_0^\pi a^2 \sin^2 \theta d\theta$$

$$= \frac{a^2}{2} \int_0^\pi \frac{1 - \cos 2\theta}{2} d\theta = \frac{a^2}{4} \int_0^\pi 1 - \cos 2\theta d\theta = \frac{a^4}{4} \left[\theta - \frac{\sin 2\theta}{2} \right]_0^\pi$$

$$= \frac{a^2}{4} \left[\pi - \frac{\sin 2\pi}{2} \right] = \frac{a^2}{4} \pi$$

Q2 $\int_0^{\pi/4} \int_0^{a \sin \theta} \frac{r}{\sqrt{a^2 - r^2}} dr d\theta$

sol

$$\int_0^{\pi/4} \int_0^{a \sin \theta} \frac{r}{\sqrt{a^2 - r^2}} dr d\theta$$

Put $a^2 - r^2 = t$

$$-2r dr = dt \Rightarrow r dr = -\frac{dt}{2}$$

$$\int_0^{\pi/4} \int_0^{a \sin \theta} \frac{-dt/\theta}{\sqrt{t}} dt d\theta$$

$$\int_0^{\pi/4} -\frac{1}{2} \int_0^{a \sin \theta} \frac{1}{t^{1/2}} dt d\theta = \int_0^{\pi/4} -\frac{1}{2} \int_0^{a \sin \theta} t^{-1/2} dt d\theta$$

$$= -\frac{1}{2} \int_0^{\pi/4} \frac{t^{1/2}}{\frac{1}{2}} \Big|_0^{a \sin \theta} d\theta = -\int_0^{\pi/4} (a^2 - t^2)^{1/2} \Big|_0^{a \sin \theta} d\theta$$

$$= -\int_0^{\pi/4} (a^2 - a^2 \sin^2 \theta)^{1/2} - (a^2 - 0) d\theta$$

$$= -\int_0^{\pi/4} a \cos \theta - a d\theta$$

$$= [a \sin \theta - a\theta]_0^{\pi/4}$$

$$= [a\theta - a \sin \theta]_0^{\pi/4} = \frac{a\pi}{4} - \frac{a}{\sqrt{2}}$$

Q) Evaluate $\iint r \sin \theta dr d\theta$ over a curve $r = a(1 - \cos \theta)$ about the initial line.

$$r = a(1 - \cos \theta)$$

$$\theta = 0 \quad \pi/4 \quad \pi/2 \quad \pi$$

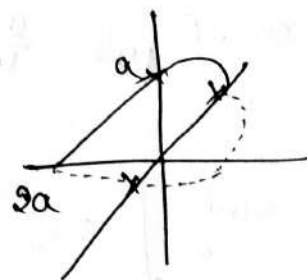
$$r = 0 \quad a(1 - \frac{1}{\sqrt{2}}) \quad a \quad 2a$$

$$\theta \rightarrow 0 \text{ to } \pi$$

$$r \rightarrow 0 \text{ to } a(1 - \cos \theta)$$

$$\iint r \sin \theta dr d\theta = \int_0^{\pi} \int_0^{a(1 - \cos \theta)} r \sin \theta dr d\theta = \int_0^{\pi} \sin \theta \left[\frac{r^2}{2} \right]_0^{a(1 - \cos \theta)} d\theta$$

$$= \frac{1}{2} \int_0^{\pi} \sin \theta a^2 (1 - \cos \theta)^2 d\theta$$



Put $1 - \cos \theta = t$

$$\frac{1}{2} \int_0^{\pi} a^2 t^2 dt = \frac{a^2}{2} \left[\frac{t^3}{3} \right]_0^{\pi} = \frac{a^2}{6} [(1 - \cos \theta)^3]_0^{\pi}$$

$$= \frac{a^2}{6} [(1 - \cos \pi)^3 - (1 - \cos 0)^3] = \frac{a^2}{6} [(1 + 1)^3 - (1 - 1)^3]$$

$$= \frac{a^2}{6} (2)^3 = \frac{8a^2}{6} = \frac{4}{3} a^2$$

Q) Find $\iint r^3 dr d\theta$ over the area included between the circle $r = 2 \sin \theta$, $r = 4 \sin \theta$

Soln

$$r = 2 \sin \theta$$

$\theta = 0$	$\pi/4$	$\pi/2$	π
$r = 0$	$\sqrt{2}$	2	0

$$r = 4 \sin \theta$$

$\theta = 0$	$\pi/4$	$\pi/2$	π
$r = 0$	$2\sqrt{2}$	4	0

$$\theta \rightarrow 0 \text{ to } \pi$$

$$r \rightarrow 2 \sin \theta \text{ to } 4 \sin \theta$$

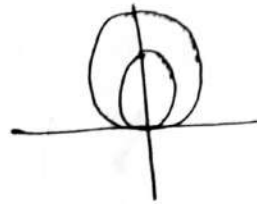
$$\iint r^3 dr d\theta = \int_0^{\pi} \int_{2 \sin \theta}^{4 \sin \theta} r^3 dr d\theta$$

$$= \int_0^{\pi} \left[\frac{r^4}{4} \right]_{2 \sin \theta}^{4 \sin \theta} d\theta$$

$$= \int_0^{\pi} \frac{1}{4} [4^4 \sin^4 \theta - 2^4 \sin^4 \theta] d\theta$$

$$= \frac{1}{4} \int_0^{\pi} 240 \sin^4 \theta d\theta = 60 \int_0^{\pi} (\sin^2 \theta)^2 d\theta = 60 \int_0^{\pi} \left[\frac{1 - \cos 2\theta}{2} \right]^2 d\theta$$

$$= \frac{60}{4} \int_0^{\pi} (1 + \cos^2 2\theta - 2 \cos 2\theta) d\theta = 15 \int_0^{\pi} 1 + \frac{1 + \cos 2\theta}{2} - 2 \cos 2\theta d\theta$$



$$= 15 \int_0^{\pi} \left[\frac{3}{2} + \frac{\cos 2\theta}{2} - 2 \cos \theta \right] d\theta$$

$$= 15 \left[\frac{3}{2} \theta + \frac{\sin 2\theta}{4} - \frac{2 \sin \theta}{2} \right]_0^{\pi} = 15 \left\{ \left(\frac{3}{2} \pi + 0 - 0 \right) - 0 \right\} = \frac{45\pi}{2}$$

change of variables:

let $x=f(u,v)$ & $y=g(u,v)$ be the relation between old variables (x,y) and new variables u,v then

$$\iint_R F(x,y) dx dy = \iint F(u,v) |J| du dv$$

for ex: If $x=r \cos \theta$ and $y=r \sin \theta$ then

$$J = \begin{vmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{vmatrix} = \begin{vmatrix} \cos \theta & -r \sin \theta \\ \sin \theta & r \cos \theta \end{vmatrix} = r$$

$$= \iint F(x,y) dx dy = \iint F(r \cos \theta, r \sin \theta) r dr d\theta$$

Q) Evaluate $\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx dy$ by changing the polar coordinates

clearly the region of integration is the first quadrant

$$\int_0^{\infty} \int_0^{\infty} e^{-(x^2+y^2)} dx dy \rightarrow \text{①}$$

taking $x=r \cos \theta$ $\Rightarrow dx dy = r dr d\theta$
 $y=r \sin \theta$

$$\left. \begin{array}{l} x^2+y^2=r^2 \\ x=0, y=0, r=0 \\ x=\infty, y=\infty, r=\infty \end{array} \right\} \begin{array}{l} \theta = \tan^{-1} \left(\frac{y}{x} \right) \\ x=0, y=0 \Rightarrow \theta = \tan^{-1} 0 \Rightarrow \theta = 0 \\ x=\infty, y=\infty \Rightarrow \theta = \frac{\pi}{2} \end{array}$$

$$\text{①} \Rightarrow \int_0^{\pi/2} \int_0^{\infty} e^{-r^2} \cdot r dr d\theta$$

$$\theta = 0, r = 0$$

Put $x^2 = t \Rightarrow 2x dx = dt$

$$\int_0^{\pi/2} \int_0^{\infty} e^{-t} \frac{dt}{2} d\theta = \frac{1}{2} \int_0^{\pi/2} e^{-t} \Big|_0^{\infty} d\theta = \frac{1}{2} \int_0^{\pi/2} -e^{-x^2} \Big|_0^{\infty} d\theta$$

$$= \frac{1}{2} \int_0^{\pi/2} [0 + 1] d\theta = \frac{1}{2} [\theta]_0^{\pi/2} = \frac{\pi}{4}$$

Q) Evaluate by transforming polar coordinates $\int_0^a \int_0^{\sqrt{a^2-x^2}} y \sqrt{x^2+y^2} dx dy$
 Sol Here region of integration is bounded by $y=0$

$$y = \sqrt{a^2 - x^2} \Rightarrow x^2 + y^2 = a^2$$

$$\theta \rightarrow 0 \text{ to } \pi/2$$

$$r \rightarrow 0 \text{ to } a$$

Taking $x = r \cos \theta$, $y = r \sin \theta$

$$dx dy = r dr d\theta$$

then $\int_0^a \int_0^{\pi/2} r \sin \theta \sqrt{r^2} r dr d\theta$

$$= \int_0^a \int_0^{\pi/2} r^3 \sin \theta d\theta dr$$

$$= \int_0^a r^3 [-\cos \theta]_0^{\pi/2} dr = \int_0^a r^3 [-\cos \frac{\pi}{2} - (-\cos 0)] dr$$

$$= \int_0^a r^3 (-1) dr = \int_0^a r^3 dr = \left[\frac{r^4}{4} \right]_0^a = \frac{a^4}{4}$$

Q) $\iint_R (x+y)^2 dx dy$ where R is the parallelogram (x,y) plane with vertices $(1,0)$ $(3,1)$ $(2,2)$ $(0,1)$ by using the

transformation $u = x + y$, $v = x - 2y$

$$u = x + y$$

$$v = x - 2y$$

① + ② $2u + v = 2x + 2y + x - 2y$

$$x = \frac{1}{3} (u+v)$$

$$u-v = x+y-x+2y$$

$$y = \frac{1}{3} (u-v)$$

$$J = \begin{vmatrix} \frac{\partial x}{\partial u} & \frac{\partial x}{\partial v} \\ \frac{\partial y}{\partial u} & \frac{\partial y}{\partial v} \end{vmatrix} = \begin{vmatrix} \frac{1}{3} & \frac{1}{3} \\ \frac{1}{3} & -\frac{1}{3} \end{vmatrix} = -\frac{1}{9} - \frac{1}{9} = -\frac{2}{9}$$

$$|J| = \frac{2}{9}$$

$$dx dy = |J| du dv = \frac{2}{9} du dv$$

limits

$$u = x+y, v = x-2y$$

$$\text{at } (1,0): u=1, v=1$$

$$\text{at } (3,1) \Rightarrow u=4, v=1$$

$$\text{at } (2,2) \Rightarrow u=4, v=-2$$

$$\text{at } (0,1) \Rightarrow u=1, v=-2$$

$$\therefore u \rightarrow 1 \text{ to } 4$$

$$v \rightarrow -2 \text{ to } 1$$

$$\int_1^4 \int_{-2}^1 \left[\frac{2}{9}u + \frac{1}{3}v + \frac{1}{3}u - \frac{1}{3}v \right] \frac{2}{9} du dv$$

$$\int_1^4 \int_{-2}^1 u^2 du dv = \frac{1}{3} \int_{-2}^1 u^2(v) du = \frac{1}{3} \int_{-2}^1 u^2(1+2v) du$$

$$= \frac{u^3}{3} \Big|_1^4 = \frac{4^3}{3} - \frac{1^3}{3} = \frac{64}{3} - \frac{1}{3} = \frac{63}{3} = 21$$

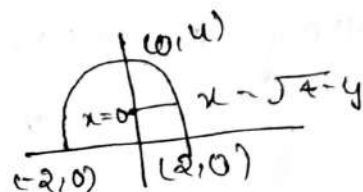
change of order of integration:

change the order of integration and evaluate

$$\int_0^2 \int_0^{4-x^2} \frac{x e^{2y}}{4-y} dy dx$$

$$\int_0^2 \int_0^{4-x^2} \frac{x e^{2y}}{4-y} dy dx \rightarrow (1)$$

The region of integration is bounded by $y=0$ (x-axis), $y=4-x^2$ which is a parabola which touches the y axis at 4 (By taking $x=0, y=4$) and x axis at -2 & 2 (By taking $y=0, x=\pm 2$)



Region of integration is 1st quadrant
 given limits of x are constant and $y=f(x)$. By changing order of integration we take limits of y as constant & x in terms of y by considering line parallel to x axis in region of integration.

$$y \rightarrow 0 \text{ to } 4$$

$$x \rightarrow 0 \text{ to } \sqrt{4-y}$$

$$\int_0^4 \int_0^{\sqrt{4-y}} \frac{x e^{2y}}{4-y} dx dy$$

$$\int_0^4 \frac{e^{2y}}{4-y} \left[\frac{x^2}{2} \right]_0^{\sqrt{4-y}} dy$$

$$= \frac{1}{2} \int_0^4 \frac{e^{2y}}{4-y} (4-y) dy$$

$$= \frac{1}{2} \int_0^4 e^{2y} dy = \frac{1}{2} \left[\frac{e^{2y}}{2} \right]_0^4$$

$$= \frac{1}{4} [e^{2(4)} - e^0] = \frac{1}{4} [e^8 - 1]$$

- Q) By changing the order of integration evaluate $\int_0^1 \int_x^{2-x} xy \, dx \, dy$
- sol) The region of integration is bounded by $y = x^2$ which is a parabola and $y = 2 - x \Rightarrow x + y = 2$ is a straight line touches x-axis at (2,0) & y-axis at (0,2) and the point of intersection is obtained by solving 2 curves as follows.



$$x + y = 2$$

$$x + x^2 = 2$$

$$x^2 + x - 2 = 0$$

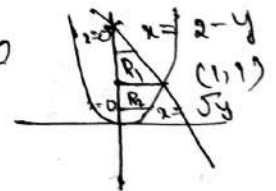
$$x = \frac{-1 \pm \sqrt{1+8}}{2} = \frac{-1 \pm 3}{2} = \frac{-1+3}{2}, \frac{-1-3}{2} = 1, -2$$

$$x = 1 \Rightarrow y = 1$$

$$x = -2 \Rightarrow y = 4$$

$\therefore$ point of intersection is (1,1) (-2,4)

By changing the order of integration we have to take the limits of y as constant & x in terms of y . By taking line parallel to x-axis in region of integration, here we observe that region is divided into two parts from the point of intersection of parabola & line.



For Region R_1 :

$$y \rightarrow 1 \text{ to } 2$$

$$x \rightarrow 0 \text{ to } 2 - y$$

$$I_1 = \int_1^2 \int_0^{2-y} xy \, dx \, dy$$

$$\begin{aligned}
 &= \int_1^2 y \frac{x^2}{2} \Big|_0^{2-y} dy = \int_1^2 \frac{y}{2} (2-y)^2 dy = \frac{1}{2} \int_1^2 y (2-y)^2 dy \\
 &= \frac{1}{2} \int_1^2 y (4+y^2-4y) dy = \frac{y^2}{2} \times 2 + \frac{y^4}{4} - \frac{2}{3} y^3 \Big|_1^2 \\
 &= \left(4 + \frac{16}{8} - \frac{2}{3} \times 8\right) - \left(1 + \frac{1}{8} - \frac{2}{3}\right) = \left(6 - \frac{16}{3}\right) - \left(\frac{9+3-16}{24}\right) = \frac{2}{3} - \frac{11}{24} = \frac{5}{24}
 \end{aligned}$$

for R_2 : $y \rightarrow 0 \text{ to } 1$ & $x \rightarrow 0 \text{ to } \sqrt{y}$

$$\begin{aligned}
 I_2 &= \int_0^1 \int_0^{\sqrt{y}} xy \, dx \, dy = \int_0^1 y \left[\frac{x^2}{2} \right]_0^{\sqrt{y}} dy = \int_0^1 \frac{y}{2} [y-0] dy = \int_0^1 \frac{y^2}{2} dy = \frac{y^3}{6} \Big|_0^1 \\
 &= \frac{1}{6}
 \end{aligned}$$

$$R_1 + R_2 = \frac{5}{24} + \frac{1}{6} = \frac{3}{8}$$

Triple Integral:

$$\begin{aligned}
 \text{Q) Evaluate } \int_0^1 \int_1^2 \int_2^3 xyz \, dx \, dy \, dz \\
 \int_0^1 \int_1^2 yz \frac{x^2}{2} \Big|_2^3 dy \, dz = \int_0^1 \int_1^2 yz \left[\frac{9}{2} - \frac{4}{2} \right] dy \, dz = \int_0^1 \int_1^2 \frac{5}{2} yz \, dy \, dz \\
 = \frac{5}{2} \int_0^1 \frac{y^2}{2} \Big|_1^2 z \, dz = \frac{5}{2} \int_0^1 \frac{1}{2} (4-1) z \, dz = \frac{5}{2} \int_0^1 \frac{3}{2} z \, dz = \frac{15}{4} \left[\frac{z^2}{2} \right]_0^1 \\
 = \frac{15}{4} \left[\frac{1}{2} - 0 \right] = \frac{15}{8}
 \end{aligned}$$

$$\begin{aligned}
 \text{Q) Evaluate } \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}} \, dz \, dy \, dx \\
 \int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^{\sqrt{1-x^2-y^2}} \frac{1}{\sqrt{1-x^2-y^2-z^2}} \, dz \, dy \, dx
 \end{aligned}$$

Put $1-x^2-y^2 = p^2$ (constant)

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \int_0^p \frac{1}{\sqrt{p^2-z^2}} \, dz \, dy \, dx = \int_0^1 \int_0^{\sqrt{1-x^2}} \sin^{-1}\left(\frac{z}{p}\right) \Big|_0^p \, dy \, dx$$

$$\int_0^1 \int_0^{\sqrt{1-x^2}} \sin^{-1}(1) - \sin^{-1}(0) dy dx = \int_0^1 \int_0^{\sqrt{1-x^2}} \frac{\pi}{2} dy dx = \frac{\pi}{2} \int_0^1 y \Big|_0^{\sqrt{1-x^2}} dx$$

$$= \frac{\pi}{2} \int_0^1 \sqrt{1-x^2} dx = \frac{\pi}{2} \left[x\sqrt{1-x^2} + \frac{1}{2} \sin^{-1}(x) \right]_0^1$$

$$= \frac{\pi}{2} \left[0 + \frac{1}{2} \sin^{-1}(1) \right] = \frac{\pi}{2} \left[\frac{1}{2} \cdot \frac{\pi}{2} \right] = \frac{\pi^2}{8}$$

Q) Evaluate $\iiint xy^2z dx dy dz$ taken over the true octant of the sphere

$$x^2 + y^2 + z^2 = a^2$$

$$x^2 + y^2 + z^2 = a^2$$

Q) $z \rightarrow 0$ to $\sqrt{a^2 - x^2 - y^2}$; $y \rightarrow 0$ to $\sqrt{a^2 - x^2}$; $x \rightarrow 0$ to a

$$\int_0^a \int_0^{\sqrt{a^2-x^2}} \int_0^{\sqrt{a^2-x^2-y^2}} xy^2z dz dy dx = \int_0^a \int_0^{\sqrt{a^2-x^2}} xy^2 \frac{z^2}{2} \Big|_0^{\sqrt{a^2-x^2-y^2}} dy dx$$

$$= \frac{1}{2} \int_0^a \int_0^{\sqrt{a^2-x^2}} xy^2(a^2 - x^2 - y^2) dy dx = \frac{1}{2} \int_0^a \int_0^{\sqrt{a^2-x^2}} (a^2xy^2 - x^3y^2 - xy^4) dy dx$$

$$= \frac{1}{2} \int_0^a a^2x \frac{y^3}{3} \Big|_0^{\sqrt{a^2-x^2}} - x^3 \frac{y^3}{3} \Big|_0^{\sqrt{a^2-x^2}} - x \frac{y^5}{5} \Big|_0^{\sqrt{a^2-x^2}} dx$$

$$= \frac{1}{2} \int_0^a \frac{a^2x}{3} (a^2-x^2)^{3/2} - \frac{x^3}{3} (a^2-x^2)^{3/2} - \frac{x}{5} (a^2-x^2)^{5/2} dx$$

$$= \frac{1}{2} \int_0^a \frac{x}{3} (a^2-x^2)^{5/2} - \frac{x}{5} (a^2-x^2)^{5/2} dx = \frac{1}{2} \int_0^a (a^2-x^2)^{5/2} \left(\frac{x}{3} - \frac{x}{5} \right) dx$$

$$= \frac{1}{2} \int_0^a (a^2-x^2)^{5/2} \left(\frac{2x}{15} \right) dx$$

$$\text{Put } a^2 - x^2 = t \Rightarrow -2x dx = dt$$

$$x dx = -dt/2$$

$$= -\frac{1}{15} \int_0^a t^{5/2} \frac{dt}{2} = -\frac{1}{30} \frac{t^{7/2}}{7/2} \Big|_0^a = -\frac{1}{15 \times 7} t^{7/2} \Big|_0^a$$

$$= -\frac{1}{105} [a^2 - x^2]^{7/2} \Big|_0^a$$

$$= -\frac{1}{105} [a^2 - a^2 - (a^2 - 0)]$$

$$= -\frac{1}{105} [-a^2] = \frac{a^2}{105}$$

Find the volume of the ellipsoid $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

volume is given by $\iiint dz dy dx$

since $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

$$z = 0 \text{ to } c \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}\right)^{1/2} ; y = 0 \text{ to } b \left(1 - \frac{x^2}{a^2}\right)^{1/2}$$

$$x = 0 \text{ to } a$$

As these are 8 octants

$$V = 8 \int_0^a \int_0^{b \left(1 - \frac{x^2}{a^2}\right)^{1/2}} \int_0^{c \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}\right)^{1/2}} dz dy dx$$

$$= 8 \int_0^a \int_0^{b \left(1 - \frac{x^2}{a^2}\right)^{1/2}} c \left(1 - \frac{x^2}{a^2} - \frac{y^2}{b^2}\right)^{1/2} dy dx$$

$$= 8 \int_0^a \int_0^{bp} \left[p^2 - \frac{y^2}{b^2}\right]^{1/2} dy dx$$

$$= \frac{8c}{b} \int_0^a \int_0^{bp} (p^2 b^2 - y^2)^{1/2} dy dx$$

Put $y = pb \sin \theta \Rightarrow dy = pb \cos \theta d\theta$

limits $\Rightarrow y = 0 \Rightarrow \theta = 0$

$y = bp \Rightarrow \theta = \pi/2$

$$= \frac{8c}{b} \int_0^a \int_0^{\pi/2} pb (1 - \sin^2 \theta)^{1/2} \cos \theta d\theta$$

$$= 8cb \int_0^a p^2 \int_0^{\pi/2} \cos^2 \theta d\theta$$

$$= 8cp \int_0^a p \int_0^{\pi/2} \frac{1 + \cos 2\theta}{2} d\theta dx$$

$$4cb \int_0^a p^2 \left\{ \theta + \frac{\sin 2\theta}{2} \right\}^{1/2} dx$$

$$= 4cb \int_0^a p^2 \left\{ \frac{\pi}{2} + 0 \right\} dx$$

$$= 2\pi cb \int_0^a 1 - \frac{x^2}{a^2} dx$$

$$= 2\pi cb \left\{ x - \frac{x^3}{a^3} \right\}_0^a$$

$$= 2\pi cb \left\{ a - \frac{a}{3} \right\}$$

$$= \frac{4\pi abc}{3}$$

change of variables in polar coordinates:

1) change of variables from cartesian to spherical polar coordinates

Here $x = r \sin \theta \cos \phi$; $y = r \sin \theta \sin \phi$; $z = r \cos \theta$

$$\text{then } \iiint f dx dy dz = \iiint f |J| dr d\theta d\phi$$

$$= \iiint f r^2 \sin \theta dr d\theta d\phi$$

2) cartesian to cylindrical coordinates

The cylindrical coordinates are given by

$$x = r \cos \theta, y = r \sin \theta, z = z$$

$$\text{Here } J = r \text{ then } \iiint dx dy dz = \iiint r dr d\theta dz$$

using spherical coordinates find the volume of the sphere

$$x^2 + y^2 + z^2 = a^2$$

we know that volume is given by $\iiint dx dy dz$

The spherical co-ordinates are

$$x = r \sin \theta \cos \phi ; y = r \sin \theta \sin \phi ; z = r \cos \theta \quad \text{and}$$

$$\iiint dx dy dz = \iiint r^2 \sin \theta dr d\theta d\phi$$

For the sphere $r \rightarrow 0$ to a

$$\theta \rightarrow 0 \text{ to } \pi$$

$$\phi = 0 \text{ to } 2\pi$$

$$V = \int_0^a \int_0^\pi \int_0^{2\pi} r^2 \sin \theta d\phi d\theta dr$$

$$= \int_0^a \int_0^\pi r^2 \sin \theta [\phi]_0^{2\pi} d\theta dr$$

$$= 2\pi \int_0^a r^2 [-\cos \theta]_0^\pi dr$$

$$= 2\pi \int_0^a r^2 [-\cos \pi + \cos 0] dr$$

$$= 4\pi \int_0^a r^2 dr$$

$$= 4\pi \left[\frac{r^3}{3} \right]_0^a$$

$$= \frac{4}{3} \pi a^3$$